PRE-ALGEBRA (8TH) · UNIT 2: EQUATIONS AND EXPRESSIONS

Phone Plan Showdown

Write and solve equations to find the best cell phone plan for any customer.

CP2

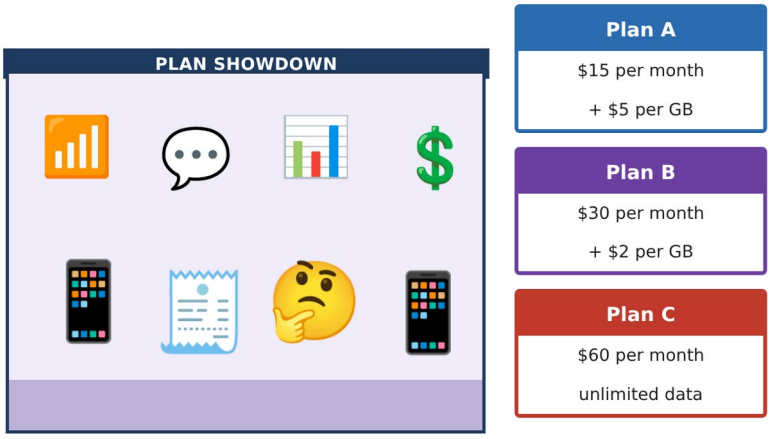
2–3 class periods

Name: _____ Date: _____ Period: _____

The Scenario

A phone store has hired you to help customers choose a plan. Each plan has a monthly fee and a price per gigabyte of data, and customers want to know which plan is cheapest for them.

You will write expressions and solve equations to compare the plans and build a guide that any customer can use.



Learning Goals

- I can write an expression for a cost that has a fixed part and a rate.
- I can solve two-step equations.
- I can solve an equation with the variable on both sides to compare two plans.

Project Timeline

When	What you will do
Day 1	Write the Costs
Day 2	Solve for the Data
Day 3	Which Plan Wins?
Final	Plan Comparison Guide + reflection

Part 1: Write the Costs

Day 1 · Let g be the number of gigabytes (GB) used in a month.

1. Write an expression for the monthly cost of Plan A and of Plan B.

Starter: *Plan A: ___ + ___g*

2. Find the cost of each plan for a customer who uses 4 GB.

3. Simplify 3(5g + 2) − 4g.

Part 2: Solve for the Data

Day 2 · Work backward from a bill.

4. A Plan A bill was $50. How many GB were used? Solve 15 + 5g = 50.

Starter: *15 + 5g = 50 → 5g = ___ → g = ___*

5. A Plan B bill was $46. How many GB were used?

6. Solve 2(x + 4) = 18.

Part 3: Which Plan Wins?

Day 3 · Compare the plans.

7. For how many GB do Plans A and B cost the same? Solve 15 + 5g = 30 + 2g. What is the cost?

8. Which plan is cheaper for 3 GB? For 8 GB?

9. At how many GB does Plan B cost the same as Plan C ($60)?

Final Product: Plan Comparison Guide

Your final product must include:

- Cost expressions for every plan
- Bills solved backward to find the data used
- Break-even points for A vs B and B vs C
- A recommendation chart: which plan for light, medium, and heavy users
- A short ad for your favorite plan

Reflection

R1. What does the solution of 15 + 5g = 30 + 2g mean for a customer?

R2. Why do you do the same thing to both sides of an equation?

Scoring Rubric

Criteria	4 – Exceeds	3 – Meets	2 – Approaching	1 – Beginning
Mathematical Accuracy	All calculations and solutions are correct; answers include units and are checked.	Nearly all work is correct; minor errors do not change conclusions.	Several errors; some conclusions are affected.	Many errors or missing work.
Reasoning & Justification	Clear, complete explanations connect every answer to the situation; uses precise vocabulary.	Explanations are mostly clear and connected to the situation.	Explanations are brief or partly unclear.	Little or no explanation.
Application & Design	Expressions and equations model the plans correctly, solutions are correct, and solution types are explained. Goes beyond the minimum with a creative, realistic choice.	Expressions and equations model the plans correctly, solutions are correct, and solution types are explained.	Design is incomplete or only partly uses the math from the unit.	Design is missing or unrelated to the unit math.
Communication & Presentation	Customer guide clearly states which plan is best for each amount of data and why. Polished, organized, and visually engaging.	Customer guide clearly states which plan is best for each amount of data and why. Organized and easy to follow.	Product is missing required parts or is hard to follow.	Product is mostly missing.